p-Adic Transfer, Signature Tails, and Homology Manifolds

From a 1980 Binghamton thesis to the 2026 Hilbert–Smith arguments

Alan J. Coppola

Ph.D. SUNY Binghamton 1979/80 · advisor L. F. McAuley

Draft for expert review · 17 Oct 2026

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models).

The ask (classical)

Can $\mathbb{Z}_p$ act continuously and faithfully on a connected finite-dimensional manifold?

Finite C_{p^k} can (lens spaces). The obstruction must use the infinite descending chain of open subgroups.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

1980: transfer

```
Over ℚ:  (1/pʳ) ∑ g*  is a projection
         ⇒  Ȟ*(X/ℤ_p; ℚ) ≅ Ȟ*(X; ℚ)^{ℤ_p}
         ⇒  no rational dimension jump

Over ℤ / 𝔽_p: transfer ∘ projection = ×pʳ ≡ 0
         ⇒  Smith–Gysin / Bockstein raises dim
         ⇒  Yang: dim_ℤ(M/ℤ_p) = n+2  (p-torsion)
```

classical Coppola 1980 · Yang · BRW

McAuley 2001 break

Orbit space dim $n+2 \Rightarrow$ saturated covers have order $> n+1$.

To force unique n-cycles he uses U^m of order $n+1$.

That cover cannot be A_p-invariant. Chain equivariance on the n-skeleton fails; the missing $(n+1)$-chain is the transfer cocycle.

new diagnosis

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

Two 2026 packagings

	OpenAI	**Dabrowski**
Shell	Newman + chart + avg	Newman + sphere control
Object	Witt lattice on S^d	L_4 signature tail
Divisibility	$4/p^k$ equal parts	single p via $\Phi_{p^a}(1) = p$
Realization	degree-1 pinch	$\mathbb{CP}^2$ germ

Homology-manifold tree

```
HM-1 non-clc / coeff mismatch     → counterexample class
HM-2 shrinking necklace            → explicit counterexample
HM-3 polyhedral L-hm               → reduces to manifold (Thm C)
HM-4 resolvable ENR / X_man ≠ ∅    → chart or X×ℝ² (Thm D)
HM-5 BFMW, I(X)≠1, X_man=∅         → crown jewel (Thms E,F)
```

Theorem C — polyhedral

Connected polyhedral L-homology n-manifold $\Rightarrow$ dense connected manifold stratum $X \setminus |K^{(n-2)}|$.

Faithful $\mathbb{Z}_p$ on $X \Rightarrow$ faithful on a topological manifold.

new packaging | relative to 2026

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

Theorem D — resolvable ENR

If $X_{\mathrm{man}} \neq \varnothing$: Newman + chart reduction.

If $I(X) = 1$: $X \times \mathbb{R}^2$ is a topological manifold (Daverman + Edwards–Quinn).

relative

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

HM-5 splits the engines

Dabrowski gap: $\Phi_{p^a}(1) = p$ only. When $p \mid I(X)$ (e.g. $I = 9$, $p = 3$), no contradiction.

OpenAI rephrase (Thm E): source $X \times S^{d-n}$, target S^d; ENR retract summands; $4c_X/p^k$ vs fixed lattice — kills every $I(X) \neq 0$.

Dabrowski fix (Thm F): k-step cyclotomic tower $\Rightarrow p^k \mid \sigma$.

Theorem G — inverse-limit no-go

```
Free step-kernel C_p on bonding fiber F_x:

Case 1  dim 0 fibers     → Cantor, lose lc⁰
Case 2  ℚ-acyclic        → Euler χ≡0 (mod p) vs χ=1
        𝔽_p-acyclic      → Smith fixed-point
Case 3  non-acyclic      → RW / Menger splicing
                         → kill local ℤ-homology

Escape: fixed points on necks → HM-2 necklace only
```

new

Lean status

- Dabrowski pin: `palomar-review/module-system-20261015` @ `b75523f`
- Toolchain: Lean `v4.35.0-rc3`, mathlib `1a547d8`
- Comparator axioms (upstream): `propext`, `Classical.choice`, `Quot.sound`
- CoppolaHS: `CyclotomicTower`, `QuinnIndex`, `PadicExclusionENR`
- *Honest:* no full ENR L-theory reimplementation in v1

What to check first

1. Theorem E (OpenAI ENR rephrase)
2. Theorem F (k-tower)
3. Theorem G (trichotomy)
4. McAuley corollary

Checklist PDF in this pack · paper Appendix B

Paper + pack

`papers/padic-homology-manifolds/main.pdf`
`presentations/padic-homology-2026/`
`lean/CoppolaHS/`
`dabrowski/upstream-palomar/` @ b75523f

Co-authorship secondary · Dabrowski natural on Theorem F

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.